

EXPERIMENT -6
RAD- (Rapid Application Development)

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Project Title: TechStore – E-commerce Web Application

Introduction

The **TechStore E-commerce Application** is a web-based platform that allows users to browse, select, and purchase electronic products such as laptops, audio devices, smartphones, and accessories.

This project follows the **Rapid Application Development (RAD)** model, focusing on quick development, continuous feedback, and iterative design.

Objective

- To design a user-friendly e-commerce interface
- To implement fast and interactive product browsing
- To simplify online purchasing with cart and checkout features
- To demonstrate RAD methodology in UI/UX development

RAD Model Overview

RAD emphasizes:

- Quick prototyping
- User involvement
- Iterative improvements
- Faster delivery

RAD Phases Applied

1. Requirement Planning

- Identified need for an online tech store
- Defined features: product listing, cart, checkout
- Target users: online shoppers

2. User Design

UI was created using Figma with the following screens:

Home Page

- Welcome banner ("Welcome to TechStore")
- Navigation bar (Home, Categories, Cart)
- "Shop Now" button

Category Section

- Laptops
- Audio
- Smartphones
- Cameras
- Wearables
- Accessories

Featured Products Page

- Product cards with:
 - Image
 - Rating
 - Description
 - Price
 - "Add to Cart" button

3. Rapid Construction

- Designed interactive components:
 - Add to cart functionality
 - Quantity adjustment (+ / -)

- Dynamic price updates
- Created structured layouts for:
 - Cart summary
 - Checkout form

Cutover (Final Stage)

- Final UI screens completed
- Ready for deployment and testing
- User-friendly navigation ensured

System Modules

User Module

- Basic navigation (no login shown but extendable)

Product Module

- Displays all available products
- Shows details like price, rating, description

Cart Module

- Add/remove items
- Increase/decrease quantity
- Displays subtotal, tax, shipping

Checkout Module

- Collects user details:
 - Name
 - Email
 - Phone
 - Address
- Shows final order summary
- “Place Order” option

Welcome to TechStore

Discover the latest in technology. Premium products at great prices.

Shop Now →

Shop by Category

View All →



Laptops



Audio



Smartphones



Cameras



Wearables



Accessories

Featured Products



★ 4.8 (266 reviews)

MacBook Pro 16"

Powerful laptop with M3 Max chip, perfect for creative professionals and developers.

\$2499

View

Add to Cart



★ 4.6 (364 reviews)

Dell XPS 15

Premium Windows laptop with stunning OLED display and powerful performance.

\$1899

View

Add to Cart



★ 4.7 (234 reviews)

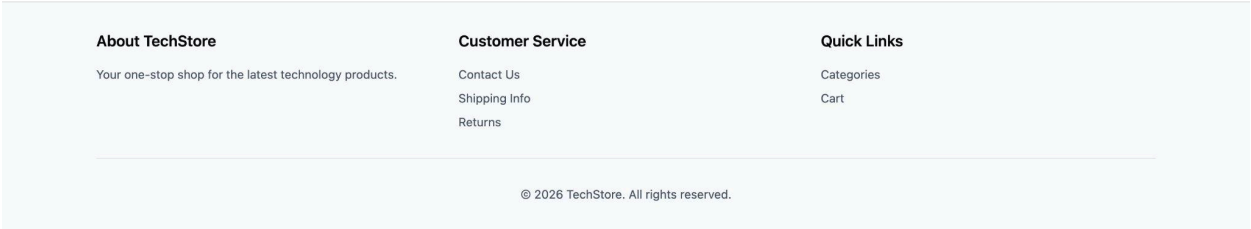
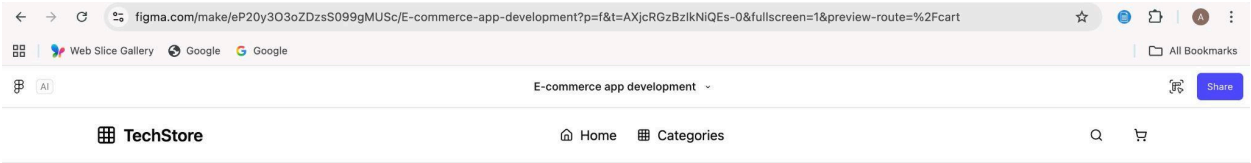
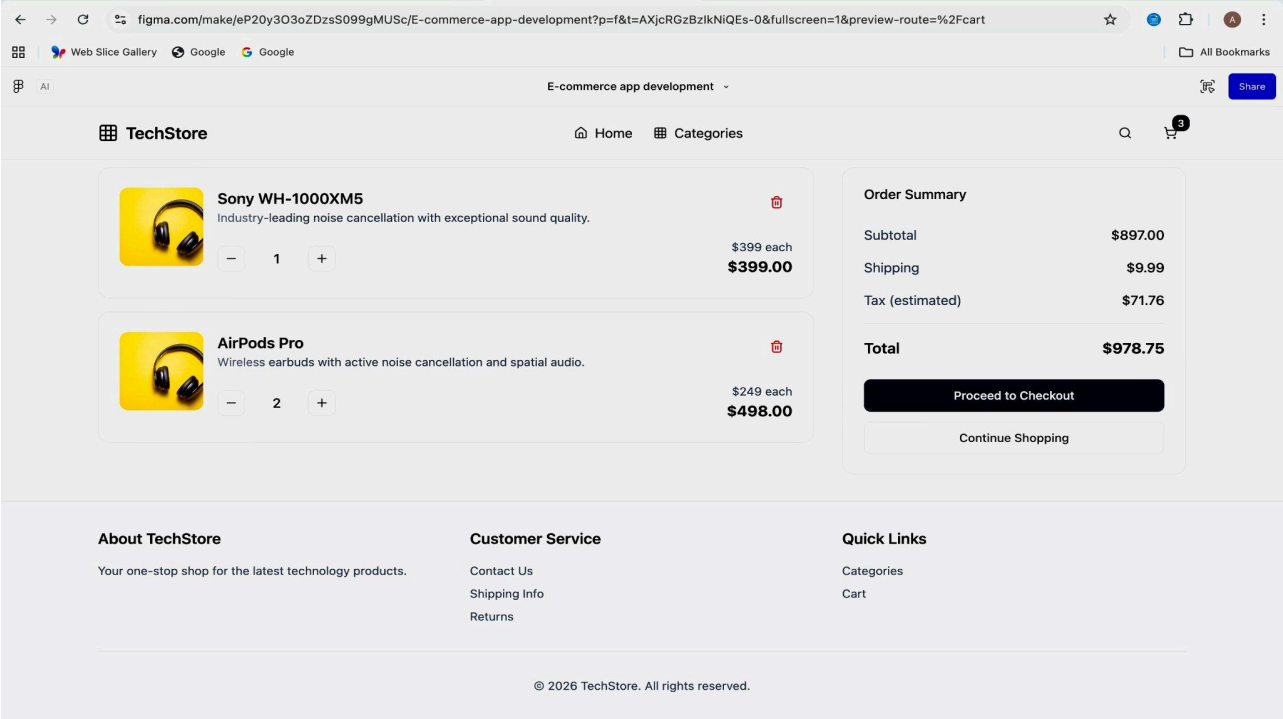
ThinkPad X1 Carbon

Lightweight business laptop with exceptional battery life and durability.

\$1599

View

Add to Cart



Workflow of the System

1. User visits homepage
2. Browses categories
3. Selects a product
4. Adds product to cart
5. Reviews cart details
6. Proceeds to checkout
7. Enters personal & shipping details
8. Places order

UI Design Highlights (From Your Screens)

- Clean and modern layout
- Gradient banner for visual appeal
- Card-based product design
- Simple navigation bar
- Responsive grid system
- Clear call-to-action buttons

Tools & Technologies

- **Design Tool:** Figma
- **Frontend:** HTML, CSS, JavaScript (conceptual)
- **Backend (optional):** Node.js / Firebase
- **Database:** MongoDB / Firestore

Advantages of the System

- Easy product browsing
- Smooth user experience
- Fast checkout process
- Scalable for future features

Future Enhancements

- User login & authentication
- Payment gateway integration
- Order tracking system
- Wishlist feature
- AI-based recommendations

Conclusion

The **TechStore E-commerce application** successfully demonstrates the **RAD model**, allowing rapid UI development with continuous improvements. The system is efficient, scalable, and user-friendly, making it suitable for real-world implementation.